

Hannah T. Nguyen

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EDUCATION

The University of Tulsa

M.S. Electrical Engineering
B.S. Electrical and Computer Engineering
Minors in Mathematics and Computer Science

Tulsa, Oklahoma
May 2028 (expected)
May 2027, GPA: 4.000

Union High School

High School Diploma

Tulsa, Oklahoma
May 2023, GPA: 4.96 (Weighted)

WORK EXPERIENCE

U.S. Government

Digital Engineering Intern

Washington D.C.

Summer 2026

- Created an enterprise-wide database tracking interface for digital forensics, streamlining legacy workflows for approvals, procurement, and renewals, improving productivity, and was presented to senior leadership.
- Designed and developed the web application using HTML, CSS, Javascript, Bootstrap, Python, and Flask, later migrating it to Node.js and Angular to improve scalability, maintainability, and performance, while using Ansible for deployment and configuration management.
- Performed hardware teardowns and comparative analysis of similar consumer electronic and electromechanical devices from multiple manufacturers, including data extraction and disassembly/reassembly of mobile, laptop, and other select hardware.
- Documented detailed technical documentation of teardowns, device comparisons, and project findings, including extensive documentation for an individual intern project.
- Communicated regularly with management to report progress, address challenges and roadblocks, and incorporate feedback for continuous improvement.

University School, University of Tulsa

Lab Assistant

Tulsa, Oklahoma

Fall 2025 – Present

- Presented hands-on projects to middle school computer science and engineering classes, promoting STEM engagement through interactive learning experiences.
- Assisted in managing a multi-node Raspberry Pi rack, supporting system setup, maintenance, and experimentation with distributed computing environments.

U.S. Government

Software Developer Intern

Washington D.C.

Summer 2025

- Developed a full-stack application from scratch to support AI-enabled triage, retrieval, and exploitation of sensitive data from complex technical collections, rapidly learning and implementing a new framework.
- Used Vue3 and Vuetify to build an intuitive user interface and AI chatbot, integrating AI API calls for efficient search and retrieval, while improving usability through structured data presentation and streamlined functionality.
- Collaborated with cross-functional teams including data scientists, engineers, and IT administrators to support AI/ML integration, cloud application development, and mission-focused data workflows.
- Improved system reliability and maintainability by debugging, cleaning, and documenting code, while communicating technical progress and issues across teams.
- Recognized for strong performance, technical capability, and ability to integrate effectively within a high-performing team.

NCFI Laboratory, University of Tulsa

Undergraduate Researcher

Tulsa, Oklahoma

Fall 2023 – May 2025

- Conducted research on the development and applications of alternative reality software and tools in order to quickly and affordably re-create digital versions of crime scenes as well as reconstruction large areas to allow for digital analysis of protection locations during event planning.

TURC Program, University of Tulsa

Volunteer Java Programming Instructor

Tulsa, Oklahoma

Summer 2024

- Co-taught two Java programming classes to 37 incoming freshmen.
- The courses gave the students a head start in their computer science education, enabling them to bypass the first two classes, Introduction to Programming and Problem Solving and Fundamentals of Algorithms and Computer Applications, and take advanced courses during their first semester in college.

Junior TURC Program, University of Tulsa
High School Researcher

Tulsa, Oklahoma
Summer 2022

- Helped update the Rayzor Hall keycard system from an Arduino to an STMicroelectronics implementation.
- Redesigned 3 PCBs with KiCad for physical adaptation of the game “Keep Talking and Nobody Explodes.”

Information Technology Department, Union Public Schools
Information Technology Intern

Tulsa, Oklahoma
Fall 2022 – Spring 2023

- Provided in-person assistance to students with information technology problems.
- Repaired laptop hardware and installed software.

SKILLS & PROFICIENCY

- **Electrical & Embedded:** Hardware reverse engineering, circuit analysis, electrical wiring, Raspberry Pi, Arduino, STM32, Altium Designer, KiCad, VHDL, STM32
- **Programming & Digital:** Python, Java, JavaScript, HTML, CSS, Vue, VHDL, MIPS, Xilinx, Git, MIPS
- **Engineering Software:** MATLAB, Simulink, Simscape, SOLIDWORKS, Fusion 360, Autodesk Inventor, VS Code, Ubuntu, L^AT_EX
- **Professional:** Project management, technical writing, product testing, independent research, leadership, Vietnamese

PROJECTS

Radio Frequency Signal Reverse Engineering Fall 2023

- Conducted a capstone research project on reverse engineering TU ID card systems, identifying three software and hardware vulnerabilities.
- Proposed and documented security solutions including password protection with lock bits, reinforced hardware design, and card number obfuscation.
- Course covers software-defined radios, radio frequency signal generation, resiliency techniques, signal analysis, reverse engineering and exploitation, radar basics, electronic warfare and cyber operations.
- Projects involve signal reversing and modulation of a radio-controlled toy, creating GNU Radio blocks, signal reversing of obfuscated communications, GPS signal exploitation, ADS-B signal exploitation and NFC signal exploitation.

Hardware Reverse Engineering Spring 2025

- Capstone project involved reverse engineering a custom, unknown PCB and implemented the functionality on an FPGA board using VHDL
- Course covers basic circuit theory, soldering, microcontrollers, field programmable gate arrays (FPGAs), serial communications, computer architectures and bootloaders.
- Projects involve circuit design, microcontroller programming, FPGAs and board-level reverse engineering.

Personal Portfolio Website

- Project involved designing and deploying a personal portfolio website using Vue3 framework hosted on Github.
- Implemented responsive layouts, interactive features, and version control with Git to showcase projects and skills.

Projectile Trajectory Simulation Fall 2025

- Project involved programming in MARS MIPS assembly and C to calculate projectile time-of-flight, maximum height, and trajectory angle based on user input.
- Applied physics equations of motion to model artillery firing scenarios, validated results across both implementations, and documented methodology, findings, and error analysis in a professional report.

Single-Port Read/Write Memory Design Spring 2025

- Lab focused on designing and constructing a 16×4-bit memory device using a 64-bit shift register, address counter, and supporting logic.
- Implemented address comparison with XNOR/AND gates, clock division using T flip-flops, and read/write functionality through multiplexers and D flip-flops.
- Verified functionality with LEDs and 7-segment displays, and debugged timing and hardware issues to ensure reliable operation.

Electrical Circuit Analysis Fall 2024

- Electrical circuit models, natural and forced response due to DC, AC and general excitation and matrix formulations and solutions of multinode circuits using Laplace transforms.

CERTIFICATIONS

Certifications

SANS GIAC Incident Handler Certification (GCIH)

January 2025

- GIAC Incident Handler (GCIH) certified professional with proven ability to detect, respond to, and resolve security incidents through deep understanding of attack techniques, vectors, tools, and effective incident response strategies.

SANS Global Industrial Cyber Security Professional Certification (GICSP)

November 2025

- Global Industrial Cyber Security Professional (GICSP) certified, bridging IT, engineering, and cybersecurity to secure industrial control systems across their full lifecycle in vendor-neutral, operational environments.

TS/SCI Clearance

AWARDS & ACTIVITIES

IEEE TU Chapter – President and Honor Society Member (HKN)

May 2024 – Present

- Led professional development workshops while engaging industry professionals for networking events.
- Increased ECE student participation in industry internship opportunities by 300%.
- Worked with the Department Chair to help improve the college.

Student Government Association – College of Engineering and Computer Science Senator September 2024 – January 2025

University of Tulsa President’s Honor Roll

January 2024 – Present

- Given to full-time students who maintain a term GPA of 4.0.

Alan Paller Cyber Scholarship

January 2024 – Present

- Prestigious scholarship for excellence in cyber studies and research, and a commitment to serve with the federal government or OPM-approved organizations. The scholarship provides a stipend and the opportunity to take SANS courses and certification exams at no cost.

University of Tulsa Presidential Scholarship

August 2023 – Present

- Full tuition scholarship awarded to high-achieving students who have demonstrated academic excellence in high school.

Oklahoma Academic Scholar

May 2023

- Honor accorded to students who accumulate a minimum grade point average of 3.7 or were in the top 10% of their graduating class and obtained a minimum of 27 ACT composite score or a 1220 SAT score.

Tulsa Engineering Scholarship

May 2023 – Present

- Scholarship awarded to high-achieving students who major in an engineering field.

FTC State Championship Control Award Sponsored by ARM, Inc.

February 2023

- Awarded to high school teams that use sensors and software to increase robot functionality. Teams must demonstrate innovative thinking to solve challenges such as autonomous operation and improving mechanical systems using sensors and intelligent control schemes.